

Important variables:

A	observational matrix with signal derivatives
A_apriori	the A apriori matrix with 0 or 1 values
A_constr	constraints given to A matrix
A_k	stacked A matrices
A_RO	observational matrix with signal derivatives for RO
Avec	the ray path vectors of each signal in tomography voxel
azi	azimuth of the satellite
BLh_pudel	coordinates of voxel edges/centers
coord	coord of intersections between RO and voxel model
dSWD	difference between observed and simulated slant delay
elev	elevation angles of ray traced signals
elevat_k	stacked elevation angle vector
epoch	processing epoch number
exPh	the values of excess phase
K	Kalman gain matrix
levels	altitude layers of tomography model
model	parameters of tomography and ray tracing models (structure)
num_lat	number of voxels in latitudinal direction
num_lev	number of voxels in vertical direction
num_lon	number of voxels in longitudinal direction
Nw_apr	apriori values of refractivities
P_apriori	the apriori weights for the tomography grid
P_constr	constraints given to P
P2	weights of observations
path_save	path to save the output matrices

Pminus	Kalman filtering predicted covariance matrix of estimation uncertainties
Q	the first guess for covariance matrix
R_SIWV	the errors of Slant Integrated Water Vapour
R_SWD	the errors of the slant delays
R_SWD_k	stacked R_SWD vector
ray	parameters of ray traced signal (structure)
ReIDist	total distance traversed by GNSS signal
rem_REC	id of receivers to delete from apriori data
rem_SAT	id of satellites to delete from apriori data
sat	id of GPS satellite
SAT_k	stacked id of GPS vector
satcoord	coordinates of the GNSS satellite
SIWV	the values of Slant Integrated Water Vapour
stat_name_k	stacked names of GNSS vector
station	structural matrix containing all informations regarding GNSS observations (structure)
station_name	names of GNSS stations
SWD	the values of Slant Delay (SD) or excess phase (EP)
SWD_apriori	the a priori values of Slant Delay (SD)
SWD_constr	constraints given to SD
SWD_k	stacked SWD vector
SWD_obs	the values of slant delays
switches	structural matrix of settings of the INTOMO processing (structure)
thin	Kalman filtering and selected tomography parameters (structure)
values	structural matrix containing inner/outer model data
xPminus	Kalman filtering predicted covariance matrix of estimation uncertainties

Important structures:

Ray:

- ray.de_lat_ray_fin.... bent ray latitude coordinates for each ray point (RP) [rad]
- ray.de_lon_ray_fin... bent ray longitude coordinates for each RP [rad]
- ray.de_alt_ray_fin.... bent ray altitude coordinates for each RP [km]
- ray.de_lat_ray_fins... straight ray latitude coordinates for each RP [rad]
- ray.de_lon_ray_fins.. straight ray longitude coordinates for each RP [rad]
- ray.de_alt_ray_fins... straight ray altitude coordinates for each RP [km]
- ray.refr..... refraction index for each RP
- ray.refh..... hydrostatic refraction index for each RP
- ray.refw..... wet refraction index for each RP
- ray.X_ray_b..... bent ray cartesian coordinates for each RP
- ray.X_ray..... straight ray cartesian coordinates for each RP
- ray.diff_dist..... last ray point distance to receiver
- ray.grad_n..... cart. grad. in each RP below upper boundary of the model
- ray.grad_ng..... spherical grad. in each RP below upper boundary of the model
- ray.t..... tangent vector in each RP below upper boundary of the model
- ray.dpa..... vert. grad. in each RP below upper boundary of the model
- ray.toc..... time of one iteration calculation
- ray.h..... 2nd grad. deriv. in each RP below upper boundary of the model
- ray.g..... vector update for ray position
- ray.de_b..... correction of initial tangent vector
- ray.dL..... final phase delay [m]
- ray.nstepb..... bent ray segment length multiplied by refractivity
- ray.step..... straight ray segment length
- ray.stepb..... bent ray segment length
- ray.i_pos..... indexes of voxels containing ray points
- ray.vox..... selected meteorological values at ray points positions
- ray.voxEM..... structure for voxels parameters without raypoints but traversed by raypath (e.g. pressure, temperature, water vapor)
- ray.voxIN..... structure for voxels parameters with raypoints (e.g. pressure, temperature, water vapor)

Warning: The structure is not saved as an output variable in default settings!

values

- Nw_apr..... apriori wet refractivity
- Nw_out..... apriori wet refractivity for outer model
- num_Nw_out..... id number of wet refractivity for outer model voxel
- Nw_obs_out..... additional wet refractivity for outer model voxel
- num_Nw..... id number of wet refractivity for inner model voxel
- Nw_obs..... additional wet refractivity for inner model voxel
- WV_apr..... apriori water vapour

- WV_out..... apriori water vapour for outer model
- WV_obs_out additional water vapour for outer model voxel
- num_WV..... id number of water vapour for inner model voxel
- WV_obs additional water vapour for inner model voxel

Switches:

- observations{'SWD'} - type of observations
- priori{'INNER','OUTER'} - apriori model
- constraints{'HORIZONTAL','VERTICAL'} - apriori model constraints %not tested
- stacking{'NO'/number} %not tested
- filter{'KALMAN'/'ROBUST'} - Kalman filtering type
- solution{'REAL'/'SYNTHETIC'} - solution type
- decorelation{'NO'/'YES'} - decorelation type
- aprModel{'DETER'/'ERA5'} - a priori model source
- amtrix{folder_name} - name for work folder to contain observation matrices
- parameterization{'constant'} - type of parameterization
- phi{'identity'} - include identity matrix
- method{'KALMAN'/'LSQ'} - Kalman or LSQ processing
- totalN{true/false} - processing in wet refractivity or (total) refractivity
- saveAtmParam{true/false} - save atmospheric apriori param (e.g. pressure) to tomography output
- coord{'Formatted'} - input format for coordinates of GNSS stations
- ROres{(resolution of RO processing [Hz])} - e.g. 1 for 50Hz, 10 for 5Hz
- refron{true/false} - ray tracing from apriori refractivities (if available)/from atmospheric parameters
- integrated{'yes'/'no'} - integrated tomography processing

station(t).	: t-number of epochs
- h(nr).	: nr-number of stations
- ZTD	: initial ZTD, gradients and conversion factor if present
- parameters	: id and coordinates of GNSS receiver [id lat lon h H]
- name	: station 4 letters abbrev
- satellite(sat)	: sat - satellite number currently 1:32
- .PRN	: PRN number
- .elevation	: satellite elevation [deg]
- .SWD	: slant wet delay
- .M_SWD	: SWD error
- .STD	: slant total delay
- .M_STD	: STD error
- .SIVW	: slant integrated water vapour
- .M_SIVW	: SIWV error
- .vmf1w	: Vienna Mapping Function wet
- .vmf1h	: Vienna Mapping Function hydrostatic
- .coord	: satellite position (ECEF)
- ro.	: radio occultation (RO) data
- .coordT	: coordinates of the transmitter (ECEF)
- .coordR	: coordinates of the receiver (ECEF)
- .date	: date of RO start

- .exL2 : RO excess phase on L2
- .exLC : RO excess phase on LC

output -

- xP estimated values of refractivities
- mxP errors of refractivities estimation

obs_set

- observations_set..... tomography processing epochs
- observations_set_SP3... SP3 processing epochs
- interpolation_set..... ray tracing processing epochs
- .all above in the format
[time_since:DOY:YEAR:GPS_WEEK:seconds_since:DOW:MM:DD:HH:MM]

model

- temp..... temperature in raytracing (RT) model nodes
- pres..... pressure in RT model nodes
- wvpr..... specific humidity in RT model nodes
- refr..... wet refractivity in RT model nodes
- refrRT..... total refractivity in RT model nodes
- temp_num_apr... temperature in RT voxel centers
- pres_num_apr... pressure refractivity in RT voxel centers
- wvpr_num_apr... specific humidity refractivity in RT voxel centers
- refr_num_apr.... wet refractivity in RT voxel centers
- refrRT_num..... total refractivity in RT voxel centers
- refr_apr..... wet refr. in tomography (TOMO) model nodes\voxel centers
- refr_aprF..... total refractivity in TOMO model nodes\voxel centers
- wvpr_apr..... specific humidity in TOMO model nodes\voxel centers
- temp_apr..... temperature in TOMO model nodes\voxel centers
- pres_apr..... pressure in TOMO model nodes\voxel centers
- BLh_pudel..... square coordinates of TOMO model nodes
- BLh_pudel_num.. square coordinates of TOMO model voxel centers
- BLh_pudel_rad... geographical coordinates of TOMO model nodes
- BLh_pudel_num_rad. geographical coordinates of TOMO model voxel centers
- BLh_outer square coordinates TOMO model nodes of outer model
- BLh_outer_num.... square coord. of TOMO model voxel centers of outer model
- BLh_outer_rad..... geographical coordinates of TOMO nodes of outer model
- BLh_outer_num_rad. geo.. coord. of TOMO model voxel centers of outer model
- levels_mid_TOMO... altitudes of TOMO model voxel centers
- num_inner..... indices of voxels in TOMO inner model
- num_outer..... indices of voxels in TOMO outer model
- rWGS..... mean Earth radius in RT model nodes
- LAT..... latitude matrix for TOMO model nodes
- LON..... longitude matrix for TOMO model nodes
- mid_levels_TOMO... altitudes of TOMO model voxel centers
- mid_lat_TOMO..... latitudes of TOMO model voxel centers

- mid_lon_TOMO..... longitudes of tomography model voxel centers

Thin

- Q..... the first guess for covariance matrix
- P2..... weights of observations
- R_SWD_k..... stacked errors of the slant delays and tomography model
- id_del..... observations deleted from the processing (see Subsec. 1...)
- theta..... log of eigenvalues from singular value decomposition
- theta_ind..... sorted theta values

Paths

Path to be set manually:

- PATH_INSTALL..... installation folder
- PATH_EXTERNALSAVE... tomography output save folder

Paths set automatically:

- pathRO..... path to folder with RO.nc files
- pathTOMO..... path for WORK directory
- pathORB..... path to SP3 data
- pathCONF..... path to configuration folder
- pathMETEO..... path to meteorological data
- pathATM..... path to ground-based data
- pathEXPORT..... tomography output save folder

Others

Amtrix variable saved as tomography output contains data for each epoch of the processing with both ray tracing data and slant delays/excess phase values. The variables included are: A, Avec, coord, dSWD, elev, not_cro (indexes of voxels not crossed by any ray path), R_SIWV, R_SWD, RelDist, SAT, SIWV, station_name, SWD and SWD_nodes_integ (integral value of SWD from apriori data)